Zebra puzzle

Zebra puzzle is a logic puzzle, also known as Einstein puzzle or Einstein's riddle. It is a puzzle in which one must deduce who owns a zebra and who drinks water given a set of 15 clues.

Color	Nationality	Pet	Drink	Cigarette
Yellow	Norwegian	Fox	Water	Kools
Blue	Danish	Horse	Tea	Chesterfield
Red	British	Snail	Milk	Winston
Green	German	Zebra	Coffee	Lucky Strike
Ivory	Spanish	Dog	Orange juice	Parliament

Rules:

1. The Norwegian lives in the first house.
2. The Norwegian lives next to the blue house.
3. The Danish drinks tea.
4. The German smokes Lucky Strike.
5. The British lives in the red house.
6. The Spanish owns the dog.
7. Kools are smoked in the yellow house.
8. Milk is drunk in the middle house.
9. The green house is immediately to the right of the ivory house.
10. Coffee is drunk in the green house.
11. The man who smokes Chesterfield lives in the house next to the man with the fox.
12. Kools are smoked in the house next to the house where the horse is kept.
13. The man who smokes Winston drinks orange juice.
14. The Norwegian lives next to the blue house.
15. The man who smokes Chesterfield lives in the house next to the man who drinks water.

SAT-based Solutions

Zebra puzzle can be formulated as a satisfiability problem. In propositional logic, it consists of 25 variables and 86 clauses. Below, we present two encodings, one by Jeremy Frank and another one by Mark Stickel. Finally, we show how to solve the puzzle using a SAT solver.

Formulation

1. Exactly one of the following holds: Yellow, Blue, Red, Green, Ivory
2. Exactly one of the following holds: Norwegian, Danish, British, German, Spanish
3. Exactly one of the following holds: Fox, Horse, Snail, Zebra, Dog
4. Exactly one of the following holds: Water, Tea, Milk, Coffee, Orange juice
5. Exactly one of the following holds: Kools, Chesterfield, Winston, Lucky Strike, Parliament
6. Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox
7. Norwegian → Yellow
8. Blue → Danish ∧ Horse
9. Danish → Tea
10. Ivory → Green
11. Green → Coffee
12. Green → Ivory
13. Lucky Strike → Orange juice
14. British → Red
15. Spanish → Dog
16. Kools → Yellow
17. Kools → Chesterfield ∨ Horse
18. Chesterfield → Fox ∨ Horse
19. Winston → Snail ∨ Zebra ∨ Dog
20. Winston → Lucky Strike ∨ Orange juice ∨ Parliament
21. Norwegian → Blue ∨ Red
22. Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red
23. Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
24. Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog

1. Formulation of the rule x→y→(x∧y) problem

Let's consider the rule x→y→(x∧y). Its CNF is (¬x∨¬y∨x∧y) ∧ (x∨y∨¬(x∧y)). Its DIMACS representation is:

```
c rule x->y->(x&y)
p cnf 4 2
-1 -2 1 2 0
1 2 -1 -2 0
```

2. Definition of variables and clauses (Jeremy Frank)

```
c Variables
c Yellow 1
c Blue 2
c Red 3
c Green 4
c Ivory 5
c Norwegian 6
c Danish 7
c British 8
c German 9
c Spanish 10
c Fox 11
c Horse 12
c Snail 13
c Zebra 14
c Dog 15
c Water 16
c Tea 17
c Milk 18
c Coffee 19
c Orange juice 20
c Kools 21
c Chesterfield 22
c Winston 23
c Lucky Strike 24
c Parliament 25
c
c Exactly one of the following holds: Yellow, Blue, Red, Green, Ivory
c Exactly one of the following holds: Norwegian, Danish, British, German, Spanish
c Exactly one of the following holds: Fox, Horse, Snail, Zebra, Dog
c Exactly one of the following holds: Water, Tea, Milk, Coffee, Orange juice
c Exactly one of the following holds: Kools, Chesterfield, Winston, Lucky Strike, Parliament
c Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox
c Norwegian → Yellow
c Blue → Danish ∧ Horse
c Danish → Tea
c Ivory → Green
c Green → Coffee
c Green → Ivory
c Lucky Strike → Orange juice
c British → Red
c Spanish → Dog
c Kools → Yellow
c Kools → Chesterfield ∨ Horse
c Chesterfield → Fox ∨ Horse
c Winston → Snail ∨ Zebra ∨ Dog
c Winston → Lucky Strike ∨ Orange juice ∨ Parliament
c Norwegian → Blue ∨ Red
c Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog
```

3. Definition of variables and clauses (Mark Stickel)

```
c Variables
c Yellow 1
c Blue 2
c Red 3
c Green 4
c Ivory 5
c Norwegian 6
c Danish 7
c British 8
c German 9
c Spanish 10
c Fox 11
c Horse 12
c Snail 13
c Zebra 14
c Dog 15
c Water 16
c Tea 17
c Milk 18
c Coffee 19
c Orange juice 20
c Kools 21
c Chesterfield 22
c Winston 23
c Lucky Strike 24
c Parliament 25
c
c Exactly one of the following holds: Yellow, Blue, Red, Green, Ivory
c Exactly one of the following holds: Norwegian, Danish, British, German, Spanish
c Exactly one of the following holds: Fox, Horse, Snail, Zebra, Dog
c Exactly one of the following holds: Water, Tea, Milk, Coffee, Orange juice
c Exactly one of the following holds: Kools, Chesterfield, Winston, Lucky Strike, Parliament
c Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox
c Norwegian → Yellow
c Blue → Danish ∧ Horse
c Danish → Tea
c Ivory → Green
c Green → Coffee
c Green → Ivory
c Lucky Strike → Orange juice
c British → Red
c Spanish → Dog
c Kools → Yellow
c Kools → Chesterfield ∨ Horse
c Chesterfield → Fox ∨ Horse
c Winston → Snail ∨ Zebra ∨ Dog
c Winston → Lucky Strike ∨ Orange juice ∨ Parliament
c Norwegian → Blue ∨ Red
c Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog
```

4. Exactly one of a set of clauses

```
c Exactly one of the following holds: Yellow, Blue, Red, Green, Ivory
c Exactly one of the following holds: Norwegian, Danish, British, German, Spanish
c Exactly one of the following holds: Fox, Horse, Snail, Zebra, Dog
c Exactly one of the following holds: Water, Tea, Milk, Coffee, Orange juice
c Exactly one of the following holds: Kools, Chesterfield, Winston, Lucky Strike, Parliament
```

5. Exactly one of a set of clauses

```
c Yellow ∨ Blue ∨ Red ∨ Green ∨ Ivory
c Norwegian ∨ Danish ∨ British ∨ German ∨ Spanish
c Fox ∨ Horse ∨ Snail ∨ Zebra ∨ Dog
c Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
c Kools ∨ Chesterfield ∨ Winston ∨ Lucky Strike ∨ Parliament
```

6. Exactly one of a set of clauses

```
c Yellow → ¬Blue ∧ ¬Red ∧ ¬Green ∧ ¬Ivory
c Blue → ¬Red ∧ ¬Green ∧ ¬Ivory
c Red → ¬Green ∧ ¬Ivory
c Green → ¬Ivory
c Norwegian → ¬Danish ∧ ¬British ∧ ¬German ∧ ¬Spanish
c Danish → ¬British ∧ ¬German ∧ ¬Spanish
c British → ¬German ∧ ¬Spanish
c German → ¬Spanish
c Fox → ¬Horse ∧ ¬Snail ∧ ¬Zebra ∧ ¬Dog
c Horse → ¬Snail ∧ ¬Zebra ∧ ¬Dog
c Snail → ¬Zebra ∧ ¬Dog
c Zebra → ¬Dog
c Water → ¬Tea ∧ ¬Milk ∧ ¬Coffee ∧ ¬Orange juice
c Tea → ¬Milk ∧ ¬Coffee ∧ ¬Orange juice
c Milk → ¬Coffee ∧ ¬Orange juice
c Coffee → ¬Orange juice
c Kools → ¬Chesterfield ∧ ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament
c Chesterfield → ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament
c Winston → ¬Lucky Strike ∧ ¬Parliament
c Lucky Strike → ¬Parliament
```

7. Norwegian → Blue

```
c Norwegian → Blue
```

8. Blue → Danish ∧ Horse

```
c Blue → Danish ∧ Horse
```

9. Danish → Tea

```
c Danish → Tea
```

10. Ivory → Green

```
c Ivory → Green
```

11. Green → Coffee

```
c Green → Coffee
```

12. Green → Ivory

```
c Green → Ivory
```

13. Lucky Strike → Orange juice

```
c Lucky Strike → Orange juice
```

14. British → Red

```
c British → Red
```

15. Spanish → Dog

```
c Spanish → Dog
```

16. Kools → Yellow

```
c Kools → Yellow
```

17. Kools → Chesterfield ∨ Horse

```
c Kools → Chesterfield ∨ Horse
```

18. Chesterfield → Fox ∨ Horse

```
c Chesterfield → Fox ∨ Horse
```

19. Winston → Snail ∨ Zebra ∨ Dog

```
c Winston → Snail ∨ Zebra ∨ Dog
```

20. Winston → Lucky Strike ∨ Orange juice ∨ Parliament

```
c Winston → Lucky Strike ∨ Orange juice ∨ Parliament
```

21. Norwegian → Blue ∨ Red

```
c Norwegian → Blue ∨ Red
```

22. Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red

```
c Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red
```

23. Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice

```
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
```

24. Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog

```
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog
```

25. Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox

```
c Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox
```

26. Yellow

```
c Yellow
```

27. Blue

```
c Blue
```

28. Red

```
c Red
```

29. Green

```
c Green
```

30. Ivory

```
c Ivory
```

31. Norwegian

```
c Norwegian
```

32. Danish

```
c Danish
```

33. British

```
c British
```

34. German

```
c German
```

35. Spanish

```
c Spanish
```

36. Fox

```
c Fox
```

37. Horse

```
c Horse
```

38. Snail

```
c Snail
```

39. Zebra

```
c Zebra
```

40. Dog

```
c Dog
```

41. Water

```
c Water
```

42. Tea

```
c Tea
```

43. Milk

```
c Milk
```

44. Coffee

```
c Coffee
```

45. Orange juice

```
c Orange juice
```

46. Kools

```
c Kools
```

47. Chesterfield

```
c Chesterfield
```

48. Winston

```
c Winston
```

49. Lucky Strike

```
c Lucky Strike
```

50. Parliament

```
c Parliament
```

51. Yellow ∨ ¬Yellow

```
c Yellow ∨ ¬Yellow
```

52. Blue ∨ ¬Blue

```
c Blue ∨ ¬Blue
```

53. Red ∨ ¬Red

```
c Red ∨ ¬Red
```

54. Green ∨ ¬Green

```
c Green ∨ ¬Green
```

55. Ivory ∨ ¬Ivory

```
c Ivory ∨ ¬Ivory
```

56. Norwegian ∨ ¬Norwegian

```
c Norwegian ∨ ¬Norwegian
```

57. Danish ∨ ¬Danish

```
c Danish ∨ ¬Danish
```

58. British ∨ ¬British

```
c British ∨ ¬British
```

59. German ∨ ¬German

```
c German ∨ ¬German
```

60. Spanish ∨ ¬Spanish

```
c Spanish ∨ ¬Spanish
```

61. Fox ∨ ¬Fox

```
c Fox ∨ ¬Fox
```

62. Horse ∨ ¬Horse

```
c Horse ∨ ¬Horse
```

63. Snail ∨ ¬Snail

```
c Snail ∨ ¬Snail
```

64. Zebra ∨ ¬Zebra

```
c Zebra ∨ ¬Zebra
```

65. Dog ∨ ¬Dog

```
c Dog ∨ ¬Dog
```

66. Water ∨ ¬Water

```
c Water ∨ ¬Water
```

67. Tea ∨ ¬Tea

```
c Tea ∨ ¬Tea
```

68. Milk ∨ ¬Milk

```
c Milk ∨ ¬Milk
```

69. Coffee ∨ ¬Coffee

```
c Coffee ∨ ¬Coffee
```

70. Orange juice ∨ ¬Orange juice

```
c Orange juice ∨ ¬Orange juice
```

71. Kools ∨ ¬Kools

```
c Kools ∨ ¬Kools
```

72. Chesterfield ∨ ¬Chesterfield

```
c Chesterfield ∨ ¬Chesterfield
```

73. Winston ∨ ¬Winston

```
c Winston ∨ ¬Winston
```

74. Lucky Strike ∨ ¬Lucky Strike

```
c Lucky Strike ∨ ¬Lucky Strike
```

75. Parliament ∨ ¬Parliament

```
c Parliament ∨ ¬Parliament
```

76. Yellow ∨ Blue ∨ Red ∨ Green ∨ Ivory

```
c Yellow ∨ Blue ∨ Red ∨ Green ∨ Ivory
```

77. Norwegian ∨ Danish ∨ British ∨ German ∨ Spanish

```
c Norwegian ∨ Danish ∨ British ∨ German ∨ Spanish
```

78. Fox ∨ Horse ∨ Snail ∨ Zebra ∨ Dog

```
c Fox ∨ Horse ∨ Snail ∨ Zebra ∨ Dog
```

79. Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice

```
c Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
```

80. Kools ∨ Chesterfield ∨ Winston ∨ Lucky Strike ∨ Parliament

```
c Kools ∨ Chesterfield ∨ Winston ∨ Lucky Strike ∨ Parliament
```

81. Yellow → ¬Blue ∧ ¬Red ∧ ¬Green ∧ ¬Ivory

```
c Yellow → ¬Blue ∧ ¬Red ∧ ¬Green ∧ ¬Ivory
```

82. Blue → ¬Red ∧ ¬Green ∧ ¬Ivory

```
c Blue → ¬Red ∧ ¬Green ∧ ¬Ivory
```

83. Red → ¬Green ∧ ¬Ivory

```
c Red → ¬Green ∧ ¬Ivory
```

84. Green → ¬Ivory

```
c Green → ¬Ivory
```

85. Norwegian → ¬Danish ∧ ¬British ∧ ¬German ∧ ¬Spanish

```
c Norwegian → ¬Danish ∧ ¬British ∧ ¬German ∧ ¬Spanish
```

86. Danish → ¬British ∧ ¬German ∧ ¬Spanish

```
c Danish → ¬British ∧ ¬German ∧ ¬Spanish
```

87. British → ¬German ∧ ¬Spanish

```
c British → ¬German ∧ ¬Spanish
```

88. German → ¬Spanish

```
c German → ¬Spanish
```

89. Fox → ¬Horse ∧ ¬Snail ∧ ¬Zebra ∧ ¬Dog

```
c Fox → ¬Horse ∧ ¬Snail ∧ ¬Zebra ∧ ¬Dog
```

90. Horse → ¬Snail ∧ ¬Zebra ∧ ¬Dog

```
c Horse → ¬Snail ∧ ¬Zebra ∧ ¬Dog
```

91. Snail → ¬Zebra ∧ ¬Dog

```
c Snail → ¬Zebra ∧ ¬Dog
```

92. Zebra → ¬Dog

```
c Zebra → ¬Dog
```

93. Water → ¬Tea ∧ ¬Milk ∧ ¬Coffee ∧ ¬Orange juice

```
c Water → ¬Tea ∧ ¬Milk ∧ ¬Coffee ∧ ¬Orange juice
```

94. Tea → ¬Milk ∧ ¬Coffee ∧ ¬Orange juice

```
c Tea → ¬Milk ∧ ¬Coffee ∧ ¬Orange juice
```

95. Milk → ¬Coffee ∧ ¬Orange juice

```
c Milk → ¬Coffee ∧ ¬Orange juice
```

96. Coffee → ¬Orange juice

```
c Coffee → ¬Orange juice
```

97. Kools → ¬Chesterfield ∧ ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament

```
c Kools → ¬Chesterfield ∧ ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament
```

98. Chesterfield → ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament

```
c Chesterfield → ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament
```

99. Winston → ¬Lucky Strike ∧ ¬Parliament

```
c Winston → ¬Lucky Strike ∧ ¬Parliament
```

100. Lucky Strike → ¬Parliament

```
c Lucky Strike → ¬Parliament
```

101. Norwegian → Blue ∨ Red

```
c Norwegian → Blue ∨ Red
```

102. Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red

```
c Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red
```

103. Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice

```
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
```

104. Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog

```
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog
```

105. Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox

```
c Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox
```

106. Yellow

```
c Yellow
```

107. Blue

```
c Blue
```

108. Red

```
c Red
```

109. Green

```
c Green
```

110. Ivory

```
c Ivory
```

111. Norwegian

```
c Norwegian
```

112. Danish

```
c Danish
```

113. British

```
c British
```

114. German

```
c German
```

115. Spanish

```
c Spanish
```

116. Fox

```
c Fox
```

117. Horse

```
c Horse
```

118. Snail

```
c Snail
```

119. Zebra

```
c Zebra
```

120. Dog

```
c Dog
```

121. Water

```
c Water
```

122. Tea

```
c Tea
```

123. Milk

```
c Milk
```

124. Coffee

```
c Coffee
```

125. Orange juice

```
c Orange juice
```

126. Kools

```
c Kools
```

127. Chesterfield

```
c Chesterfield
```

128. Winston

```
c Winston
```

129. Lucky Strike

```
c Lucky Strike
```

130. Parliament

```
c Parliament
```

1. Formulation of the rule x→y→(x∧y) problem

Let's consider the rule x→y→(x∧y). Its CNF is (¬x∨¬y∨x∧y) ∧ (x∨y∨¬(x∧y)). Its DIMACS representation is:

```
p cnf 4 2
-1 -2 1 2 0
1 2 -1 -2 0
```

2. Definition of variables and clauses (Jeremy Frank)

```
c Variables
c Yellow 1
c Blue 2
c Red 3
c Green 4
c Ivory 5
c Norwegian 6
c Danish 7
c British 8
c German 9
c Spanish 10
c Fox 11
c Horse 12
c Snail 13
c Zebra 14
c Dog 15
c Water 16
c Tea 17
c Milk 18
c Coffee 19
c Orange juice 20
c Kools 21
c Chesterfield 22
c Winston 23
c Lucky Strike 24
c Parliament 25
c
c Exactly one of the following holds: Yellow, Blue, Red, Green, Ivory
c Exactly one of the following holds: Norwegian, Danish, British, German, Spanish
c Exactly one of the following holds: Fox, Horse, Snail, Zebra, Dog
c Exactly one of the following holds: Water, Tea, Milk, Coffee, Orange juice
c Exactly one of the following holds: Kools, Chesterfield, Winston, Lucky Strike, Parliament
c Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox
c Norwegian → Yellow
c Blue → Danish ∧ Horse
c Danish → Tea
c Ivory → Green
c Green → Coffee
c Green → Ivory
c Lucky Strike → Orange juice
c British → Red
c Spanish → Dog
c Kools → Yellow
c Kools → Chesterfield ∨ Horse
c Chesterfield → Fox ∨ Horse
c Winston → Snail ∨ Zebra ∨ Dog
c Winston → Lucky Strike ∨ Orange juice ∨ Parliament
c Norwegian → Blue ∨ Red
c Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog
```

3. Definition of variables and clauses (Mark Stickel)

```
c Variables
c Yellow 1
c Blue 2
c Red 3
c Green 4
c Ivory 5
c Norwegian 6
c Danish 7
c British 8
c German 9
c Spanish 10
c Fox 11
c Horse 12
c Snail 13
c Zebra 14
c Dog 15
c Water 16
c Tea 17
c Milk 18
c Coffee 19
c Orange juice 20
c Kools 21
c Chesterfield 22
c Winston 23
c Lucky Strike 24
c Parliament 25
c
c Exactly one of the following holds: Yellow, Blue, Red, Green, Ivory
c Exactly one of the following holds: Norwegian, Danish, British, German, Spanish
c Exactly one of the following holds: Fox, Horse, Snail, Zebra, Dog
c Exactly one of the following holds: Water, Tea, Milk, Coffee, Orange juice
c Exactly one of the following holds: Kools, Chesterfield, Winston, Lucky Strike, Parliament
c Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox
c Norwegian → Yellow
c Blue → Danish ∧ Horse
c Danish → Tea
c Ivory → Green
c Green → Coffee
c Green → Ivory
c Lucky Strike → Orange juice
c British → Red
c Spanish → Dog
c Kools → Yellow
c Kools → Chesterfield ∨ Horse
c Chesterfield → Fox ∨ Horse
c Winston → Snail ∨ Zebra ∨ Dog
c Winston → Lucky Strike ∨ Orange juice ∨ Parliament
c Norwegian → Blue ∨ Red
c Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog
```

4. Exactly one of a set of clauses

```
c Exactly one of the following holds: Yellow, Blue, Red, Green, Ivory
c Exactly one of the following holds: Norwegian, Danish, British, German, Spanish
c Exactly one of the following holds: Fox, Horse, Snail, Zebra, Dog
c Exactly one of the following holds: Water, Tea, Milk, Coffee, Orange juice
c Exactly one of the following holds: Kools, Chesterfield, Winston, Lucky Strike, Parliament
```

5. Exactly one of a set of clauses

```
c Yellow ∨ Blue ∨ Red ∨ Green ∨ Ivory
c Norwegian ∨ Danish ∨ British ∨ German ∨ Spanish
c Fox ∨ Horse ∨ Snail ∨ Zebra ∨ Dog
c Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
c Kools ∨ Chesterfield ∨ Winston ∨ Lucky Strike ∨ Parliament
```

6. Exactly one of a set of clauses

```
c Yellow → ¬Blue ∧ ¬Red ∧ ¬Green ∧ ¬Ivory
c Blue → ¬Red ∧ ¬Green ∧ ¬Ivory
c Red → ¬Green ∧ ¬Ivory
c Green → ¬Ivory
c Norwegian → ¬Danish ∧ ¬British ∧ ¬German ∧ ¬Spanish
c Danish → ¬British ∧ ¬German ∧ ¬Spanish
c British → ¬German ∧ ¬Spanish
c German → ¬Spanish
c Fox → ¬Horse ∧ ¬Snail ∧ ¬Zebra ∧ ¬Dog
c Horse → ¬Snail ∧ ¬Zebra ∧ ¬Dog
c Snail → ¬Zebra ∧ ¬Dog
c Zebra → ¬Dog
c Water → ¬Tea ∧ ¬Milk ∧ ¬Coffee ∧ ¬Orange juice
c Tea → ¬Milk ∧ ¬Coffee ∧ ¬Orange juice
c Milk → ¬Coffee ∧ ¬Orange juice
c Coffee → ¬Orange juice
c Kools → ¬Chesterfield ∧ ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament
c Chesterfield → ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament
c Winston → ¬Lucky Strike ∧ ¬Parliament
c Lucky Strike → ¬Parliament
```

7. Norwegian → Blue

```
c Norwegian → Blue
```

8. Blue → Danish ∧ Horse

```
c Blue → Danish ∧ Horse
```

9. Danish → Tea

```
c Danish → Tea
```

10. Ivory → Green

```
c Ivory → Green
```

11. Green → Coffee

```
c Green → Coffee
```

12. Green → Ivory

```
c Green → Ivory
```

13. Lucky Strike → Orange juice

```
c Lucky Strike → Orange juice
```

14. British → Red

```
c British → Red
```

15. Spanish → Dog

```
c Spanish → Dog
```

16. Kools → Yellow

```
c Kools → Yellow
```

17. Kools → Chesterfield ∨ Horse

```
c Kools → Chesterfield ∨ Horse
```

18. Chesterfield → Fox ∨ Horse

```
c Chesterfield → Fox ∨ Horse
```

19. Winston → Snail ∨ Zebra ∨ Dog

```
c Winston → Snail ∨ Zebra ∨ Dog
```

20. Winston → Lucky Strike ∨ Orange juice ∨ Parliament

```
c Winston → Lucky Strike ∨ Orange juice ∨ Parliament
```

21. Norwegian → Blue ∨ Red

```
c Norwegian → Blue ∨ Red
```

22. Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red

```
c Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red
```

23. Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice

```
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
```

24. Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog

```
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog
```

25. Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox

```
c Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox
```

26. Yellow

```
c Yellow
```

27. Blue

```
c Blue
```

28. Red

```
c Red
```

29. Green

```
c Green
```

30. Ivory

```
c Ivory
```

31. Norwegian

```
c Norwegian
```

32. Danish

```
c Danish
```

33. British

```
c British
```

34. German

```
c German
```

35. Spanish

```
c Spanish
```

36. Fox

```
c Fox
```

37. Horse

```
c Horse
```

38. Snail

```
c Snail
```

39. Zebra

```
c Zebra
```

40. Dog

```
c Dog
```

41. Water

```
c Water
```

42. Tea

```
c Tea
```

43. Milk

```
c Milk
```

44. Coffee

```
c Coffee
```

45. Orange juice

```
c Orange juice
```

46. Kools

```
c Kools
```

47. Chesterfield

```
c Chesterfield
```

48. Winston

```
c Winston
```

49. Lucky Strike

```
c Lucky Strike
```

50. Parliament

```
c Parliament
```

51. Yellow ∨ ¬Yellow

```
c Yellow ∨ ¬Yellow
```

52. Blue ∨ ¬Blue

```
c Blue ∨ ¬Blue
```

53. Red ∨ ¬Red

```
c Red ∨ ¬Red
```

54. Green ∨ ¬Green

```
c Green ∨ ¬Green
```

55. Ivory ∨ ¬Ivory

```
c Ivory ∨ ¬Ivory
```

56. Norwegian ∨ ¬Norwegian

```
c Norwegian ∨ ¬Norwegian
```

57. Danish ∨ ¬Danish

```
c Danish ∨ ¬Danish
```

58. British ∨ ¬British

```
c British ∨ ¬British
```

59. German ∨ ¬German

```
c German ∨ ¬German
```

60. Spanish ∨ ¬Spanish

```
c Spanish ∨ ¬Spanish
```

61. Fox ∨ ¬Fox

```
c Fox ∨ ¬Fox
```

62. Horse ∨ ¬Horse

```
c Horse ∨ ¬Horse
```

63. Snail ∨ ¬Snail

```
c Snail ∨ ¬Snail
```

64. Zebra ∨ ¬Zebra

```
c Zebra ∨ ¬Zebra
```

65. Dog ∨ ¬Dog

```
c Dog ∨ ¬Dog
```

66. Water ∨ ¬Water

```
c Water ∨ ¬Water
```

67. Tea ∨ ¬Tea

```
c Tea ∨ ¬Tea
```

68. Milk ∨ ¬Milk

```
c Milk ∨ ¬Milk
```

69. Coffee ∨ ¬Coffee

```
c Coffee ∨ ¬Coffee
```

70. Orange juice ∨ ¬Orange juice

```
c Orange juice ∨ ¬Orange juice
```

71. Kools ∨ ¬Kools

```
c Kools ∨ ¬Kools
```

72. Chesterfield ∨ ¬Chesterfield

```
c Chesterfield ∨ ¬Chesterfield
```

73. Winston ∨ ¬Winston

```
c Winston ∨ ¬Winston
```

74. Lucky Strike ∨ ¬Lucky Strike

```
c Lucky Strike ∨ ¬Lucky Strike
```

75. Parliament ∨ ¬Parliament

```
c Parliament ∨ ¬Parliament
```

76. Yellow ∨ Blue ∨ Red ∨ Green ∨ Ivory

```
c Yellow ∨ Blue ∨ Red ∨ Green ∨ Ivory
```

77. Norwegian ∨ Danish ∨ British ∨ German ∨ Spanish

```
c Norwegian ∨ Danish ∨ British ∨ German ∨ Spanish
```

78. Fox ∨ Horse ∨ Snail ∨ Zebra ∨ Dog

```
c Fox ∨ Horse ∨ Snail ∨ Zebra ∨ Dog
```

79. Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice

```
c Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
```

80. Kools ∨ Chesterfield ∨ Winston ∨ Lucky Strike ∨ Parliament

```
c Kools ∨ Chesterfield ∨ Winston ∨ Lucky Strike ∨ Parliament
```

81. Yellow → ¬Blue ∧ ¬Red ∧ ¬Green ∧ ¬Ivory

```
c Yellow → ¬Blue ∧ ¬Red ∧ ¬Green ∧ ¬Ivory
```

82. Blue → ¬Red ∧ ¬Green ∧ ¬Ivory

```
c Blue → ¬Red ∧ ¬Green ∧ ¬Ivory
```

83. Red → ¬Green ∧ ¬Ivory

```
c Red → ¬Green ∧ ¬Ivory
```

84. Green → ¬Ivory

```
c Green → ¬Ivory
```

85. Norwegian → ¬Danish ∧ ¬British ∧ ¬German ∧ ¬Spanish

```
c Norwegian → ¬Danish ∧ ¬British ∧ ¬German ∧ ¬Spanish
```

86. Danish → ¬British ∧ ¬German ∧ ¬Spanish

```
c Danish → ¬British ∧ ¬German ∧ ¬Spanish
```

87. British → ¬German ∧ ¬Spanish

```
c British → ¬German ∧ ¬Spanish
```

88. German → ¬Spanish

```
c German → ¬Spanish
```

89. Fox → ¬Horse ∧ ¬Snail ∧ ¬Zebra ∧ ¬Dog

```
c Fox → ¬Horse ∧ ¬Snail ∧ ¬Zebra ∧ ¬Dog
```

90. Horse → ¬Snail ∧ ¬Zebra ∧ ¬Dog

```
c Horse → ¬Snail ∧ ¬Zebra ∧ ¬Dog
```

91. Snail → ¬Zebra ∧ ¬Dog

```
c Snail → ¬Zebra ∧ ¬Dog
```

92. Zebra → ¬Dog

```
c Zebra → ¬Dog
```

93. Water → ¬Tea ∧ ¬Milk ∧ ¬Coffee ∧ ¬Orange juice

```
c Water → ¬Tea ∧ ¬Milk ∧ ¬Coffee ∧ ¬Orange juice
```

94. Tea → ¬Milk ∧ ¬Coffee ∧ ¬Orange juice

```
c Tea → ¬Milk ∧ ¬Coffee ∧ ¬Orange juice
```

95. Milk → ¬Coffee ∧ ¬Orange juice

```
c Milk → ¬Coffee ∧ ¬Orange juice
```

96. Coffee → ¬Orange juice

```
c Coffee → ¬Orange juice
```

97. Kools → ¬Chesterfield ∧ ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament

```
c Kools → ¬Chesterfield ∧ ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament
```

98. Chesterfield → ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament

```
c Chesterfield → ¬Winston ∧ ¬Lucky Strike ∧ ¬Parliament
```

99. Winston → ¬Lucky Strike ∧ ¬Parliament

```
c Winston → ¬Lucky Strike ∧ ¬Parliament
```

100. Lucky Strike → ¬Parliament

```
c Lucky Strike → ¬Parliament
```

101. Norwegian → Blue ∨ Red

```
c Norwegian → Blue ∨ Red
```

102. Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red

```
c Norwegian → Blue ∨ Green ∨ Ivory ∨ Yellow ∨ Red
```

103. Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice

```
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice
```

104. Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog

```
c Chesterfield → Water ∨ Tea ∨ Milk ∨ Coffee ∨ Orange juice ∨ Snail ∨ Zebra ∨ Dog
```

105. Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox

```
c Norwegian ∧ Blue ∧ Milk ∧ Kools ∧ Fox
```

106. Yellow

```
c Yellow
```

107. Blue

```
c Blue
```

108. Red

```
c Red
```

109. Green

```
c Green
```

110. Ivory

```
c
```